2 **(a)** Fig. 2.1 shows the variation of the gravitational potential ϕ with distance r along a line joining the centres of the moon and the Earth.

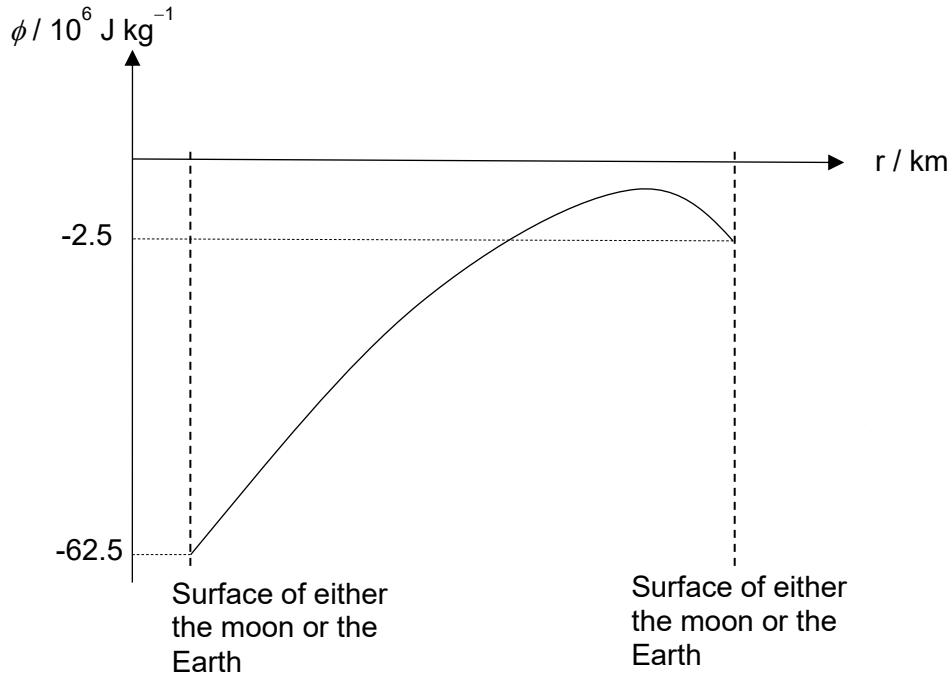

Fig. 2.1

(i) Explain whether r is measured from the moon or the Earth.

.....

..... [1]

(ii) A 1500 kg meteorite needs a minimum energy of 2.25 x 10^9 J to reach the neutral point as it travels from the moon to the Earth.

1. Explain what is meant by *the neutral point between the moon and the Earth*.

.....

..... [1]

2. Determine the gravitational potential at the neutral point.

gravitational potential = ……. J kg^{-1} [2]

3. Determine the speed at which the meteorite would hit the Earth. You may neglect atmospheric resistance.

speed = …… m s^{-1} [2]

(b) The mass of the Earth is 5.98×10^{24} kg and the moon takes 27.4 days to orbit the Earth.

(i) Show that the distance between the centre of the Earth and the moon is about 384 000 km. [2]

(ii) Hence, determine the linear speed of the moon.

linear speed = …….. m s^{-1} [2]

(c) An astronaut in a spacecraft orbits around the Earth in a circular path. The astronaut appears to float inside the spacecraft. Students A and B provide some explanations about this observation of weightlessness.

1. Student A claims that the astronaut floats because he has no weight.
2. Student B claims that the astronaut floats because both the astronaut and the spacecraft have the same centripetal force.

Comment on the validity of each of these explanations.

1. .....

.....

2. .....

..... [2]